Attributing Training Data Interactions via Influence Functions

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Research Meeting

2026-03-19

Recap: Our Contribution — An Interaction Attribution Term

Apply second-order Taylor expansion to response function f:

$$f(\theta_\varepsilon) - f(\theta^*) \approx \underbrace{-\varepsilon\, \mathbf{g}_f^\top \bar{\mathbf{v}}}_{I_1} + \frac{\varepsilon^2}{2}\Big[\underbrace{\bar{\mathbf{v}}^\top H_f \bar{\mathbf{v}}}_{T_1 \text{ (ours)}} + \underbrace{2\,\mathbf{u}^\top H_{\mathcal{S}}\, \bar{\mathbf{v}}}_{T_2 \text{ (Basu et al.)}}\Big]$$

T_1 is an **interaction attribution term**:

$$T_1 = \frac{1}{|\mathcal{S}|^2} \sum_{i,j \in \mathcal{S}} \mathbf{v}_i^\top H_f \mathbf{v}_j$$

Each term $\mathbf{v}_i^\top H_f \mathbf{v}_j$ measures how data point i and j *interact* under the model's response curvature.

Updates Since Last Meeting

Group Influence Prediction

- Added **CIFAR-10 / ResNet-9** (EK-FAC) — first CNN setting
- Added baselines: **TRAK** and **TracIn**

Greedy Subset Selection

- Iteratively remove training data to maximize test performance
- I_1 vs $I_1 + T_1$: does the interaction term improve data selection?

Group Influence Prediction: Experiment Setup

Task: Predict group influence on similar-data subsets (logit-space nearest neighbors).

Dataset	Model	Hessian	$\lvert\mathcal{S}\rvert$	n_{subsets}	**New**
MNIST	LR	Exact	400	50	
MNIST	MLP	EK-FAC	400	50	
FashionMNIST	MLP	EK-FAC	400	50	
Concrete	MLP	EK-FAC	200	50	
Parkinsons	MLP	EK-FAC	400	50	
CIFAR-10	ResNet-9	EK-FAC	400	50	✓

Group Influence Prediction: Baselines

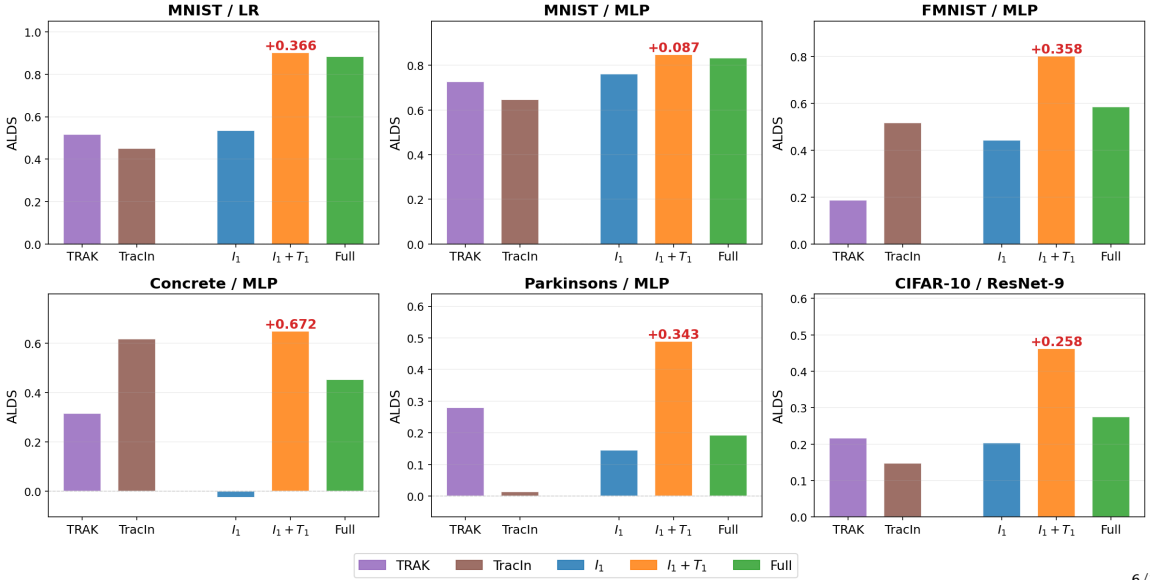
All baselines are **additive** (no interaction term).

All three predict group influence as an inner product between transformed $\mathbf{g}_f$ and $\mathbf{g}_{\mathcal{S}}$:

- I_1[1]: $\mathbf{g}_f^\top H^{-1} \mathbf{g}_{\mathcal{S}}$
- **TRAK**[2]: $(P\mathbf{g}_f)^\top P\, \mathbf{g}_{\mathcal{S}}$
- **TracIn**[3]: $\sum_c \eta_c\, \mathbf{g}_f^{(c)\top} \mathbf{g}_{\mathcal{S}}^{(c)}$ — $\mathbf{g}^{(c)}$: gradients at training step c

[1] Koh & Liang, "Understanding Black-box Predictions via Influence Functions," ICML 2017
[2] Park et al., "TRAK: Attributing Model Behavior at Scale," ICML 2023
[3] Pruthi et al., "Estimating Training Data Influence by Tracing Gradient Descent," NeurIPS 2020

Group Influence Prediction: Results



Greedy Subset Selection: Setup

Task: Iteratively remove training data predicted to be least beneficial.

I_1 **only (no interaction):**

- Removal decisions are independent of which points have already been removed

$I_1 + T_1$ **(with interaction):**

- T_1 accounts for interactions when selecting which point to remove
- Removing similar points reduces T_1 less $\Rightarrow$ greedy prefers removing from **diverse** directions

Greedy Subset Selection: Efficient Iterative T_1 Update

Naively recomputing T_1 at each step is $O(|\mathcal{S}|^2)$. Instead, use a **recurrence relation**:

$$\underbrace{\sum_{i,j\in\mathcal{S}_{k+1}} \mathbf{w}_i^\top \mathbf{v}_j}_{T_1^{\mathcal{S}_{k+1}}} = \underbrace{\sum_{i,j\in\mathcal{S}_k} \mathbf{w}_i^\top \mathbf{v}_j}_{T_1^{\mathcal{S}_k}} + 2\sum_{i\in\mathcal{S}_k} \mathbf{w}_i^\top \mathbf{v}_{k+1} + \mathbf{w}_{k+1}^\top \mathbf{v}_{k+1}$$

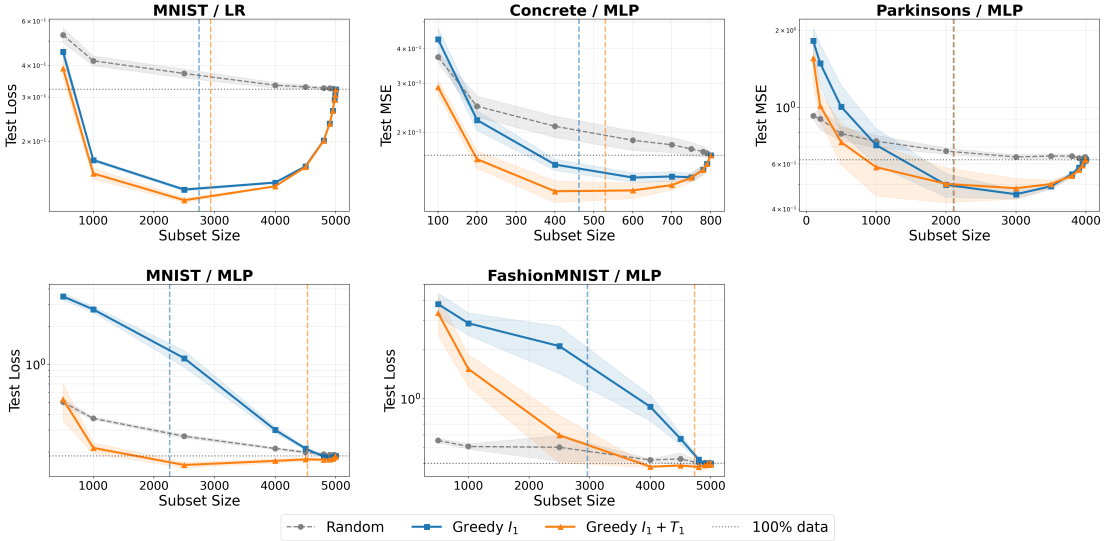
where $\mathbf{w}_i = H_f \mathbf{v}_i$.

Precomputation:

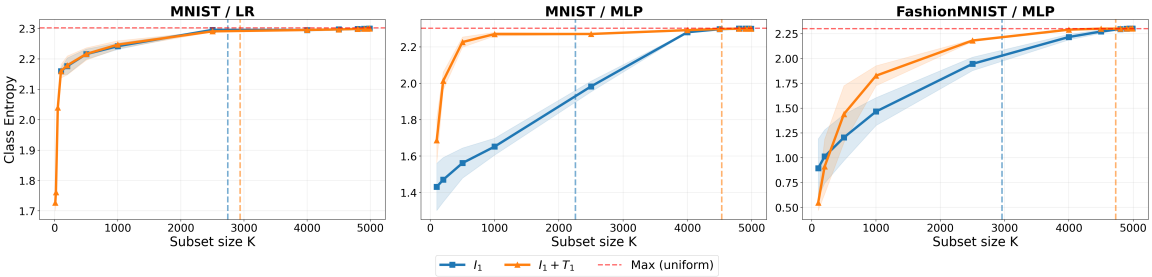
- $\mathbf{v}_i = H^{-1}\nabla_\theta \ell(z_i, \hat{\theta})$ for all training points
- $\mathbf{w}_i = H_f \mathbf{v}_i$ for all training points

$\Rightarrow$ Each greedy step reduces to **vector inner products**.

Greedy Subset Selection: Results



Greedy Subset Selection: Class Distribution Entropy



- $I_1 + T_1$ maintains higher class entropy (more uniform distribution)
- Interaction term acts as a **diversity regularizer**

Next Steps

- Multi-seed group influence prediction
- Wikitext2 / GPT-2 experiments
- TRAK / TracIn baselines for greedy subset selection
- Theoretical analysis of the diversity–helpfulness tradeoff in T_1